Liam Tanner

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Technical Skills

Languages: Python, C++, C, Java, SQL, Bash
Backend & Systems: Linux, FastAPI, RESTful APIs, PostgreSQL, SQLite, MongoDB, ROS 2
Cloud & DevOps: Docker, Git, GitHub Actions, CI/CD, AWS (Cloud Practitioner Certified)
ML & Compute: PyTorch, Scikit-learn, LLM/RAG Pipelines, ARM NEON SIMD

Education

University of Victoria Victoria, BC
Bachelor of Software Engineering *Sep 2021 to Aug 2026*
- 4.00/4.00 GPA - 93% Avg (2025/2026 Session) | Courses: Optimization for ML, Computer Vision, Data Mining

Experience

Software Developer Jun 2025 to Dec 2025
Natural Resources Canada *Victoria, BC*
- Designed and automated end-to-end ETL workflows to transform PostgreSQL datasets into SQLite and flat-file releases, reducing processing time from hours to seconds.
- Developed hundreds of automated JMeter and Selenium tests across more than 60 pages, replacing repetitive manual verification with automated regression testing and catching issues before deployment.
- Rebuilt a legacy public website without access to its original source and authored Linux Bash scripts to automate extraction, repair, and repeatable deployment on Apache.
- Developed a release-versioning system using SQL functions, triggers, and encryption to encode release metadata, archive historical versions, and support future data-sharing workflows.

Software Engineer Jan 2025 to Aug 2026
AUVIC - Autonomous Underwater Vehicles Club *Victoria, BC*
- Built a ROS 2 runtime configuration pipeline in C++ and YAML for PID and Kalman filter parameters, eliminating recompilation downtime and enabling real-time tuning during pool testing.
- Diagnosed and resolved thruster saturation and control issues during pool testing, contributing to the team's advancement to the RoboSub 2026 semi-finals.

Research Assistant - Induced Seismicity Project Sep 2023 to Apr 2024
Natural Resources Canada *Sidney, BC*
- Developed three increasingly complex CO_2 injection simulations using Python-generated meshes and the OpenGeoSys finite element solver, becoming the first NRCan researcher to successfully implement the workflow.
- Advanced the project to a real-world case study by building a 3-layer finite element simulation of the SECURE INGA disposal well using precise depth profiles and material properties.

Projects

AI Incident Orchestrator | *Python, PostgreSQL, LLMs, RAG* May 2026 to Aug 2026
- Designed and implemented the backend architecture for an LLM/RAG orchestration pipeline in Python, coordinating incident context retrieval and Groq API calls to diagnose cybersecurity incidents and generate mitigation recommendations.
- Built a PostgreSQL database client and data-access layer to retrieve incident context across the orchestration pipeline, separating database logic from the application layer.

EfficientAD Anomaly Detector | *Python, PyTorch, Computer Vision* Mar 2026
- Implemented the EfficientAD anomaly detection pipeline in PyTorch, including the Patch Description Network and teacher/student/autoencoder training workflow.
- Identified and removed training bottlenecks to complete 70,000 iterations in under 2 hours, achieving 0.88 AUROC with sub-millisecond inference latency.

Video Motion Estimation Optimizations | *C, ARM Assembly, SIMD* Jul 2026 to Aug 2026
- Optimized a 16x16 Sum-of-Absolute-Differences kernel for video motion estimation on ARM using a custom SAD4 instruction, patched QEMU emulation, and inline assembly.
- Achieved an 11.89x execution speedup with the custom operation and a peak 34.93x speedup using ARM NEON SIMD vectorization.